

CEGEP VANIER COLLEGE
Department of Computer Science
420-231-VA Database

Team Project- Deliverable #3: Query Development

Deliverable 3 Due: May 11, 2026

Teacher: Nagat Drawel

Instructions

1. **Your deliverable must contain the following:**
 - a. The SQL statements for each of the defined queries **and its corresponding execution results.**
 - b. Final Report (**pdf format**)
2. **Each team will give a short presentation.** The presentation is not intended to be a formal PowerPoint presentation. Instead, each team will briefly showcase their database design, emphasizing the work done by each member of the team. There will be a short question and answer session following each presentation. The presentations will be held during the class hours (May 12 and May 14).

Query Development

1. Develop at least 3 complex queries and 2 views. SELECT queries that represent answers to likely business questions to be faced by the users of your database system.
 - a. For each of these queries, please also provide a 1-2 sentence, for why that query is relevant and interesting to the owners of the system.
 - b. Motivate and explain at least two views, with at least part of each rationale being related to security, or the principle of “need to know” (e.g., a “user” only needs access to information concerning them).
 - c. Your queries should not be simple (i.e., you should not just list everything from your TOUR table!).
 - d. In the queries and views, you must demonstrate the uses of aggregate operators, group by clause, order by clause, subqueries and involve table joins
2. Create 2 stored procedures to enforce essential business rules within the database system. These procedures should enable users to perform actions such as insertion, deletion, or updating via the stored procedures. Ensure that the stored procedures incorporate parameters, employ conditional logic, call a user-defined function, and include a TRY...CATCH block. Additionally, provide a clear explanation of the intended purpose for each stored procedure

3. Write a database trigger that ensures when a new billing record is inserted into the Billing table; the corresponding **pat_balance** in the Patient table is automatically updated. Specifically, the value in the **bill_balance** column of the inserted record must be added to the existing balance of the associated patient.

You may need to modify your schema to support this requirement, such as adding a **pat_balance** column to the PATIENT table. Moreover, if there is no direct relationship between the BILL and PATIENT tables, your trigger must determine the correct patient by traversing the existing relationships in the database.

Additionally, your trigger should account for scenarios where the operation does not meet the required business rules, such as when the referenced patient does not exist, the billing amount is invalid (e.g., negative or NULL), or data integrity constraints are violated.

In the case of an error, the trigger must raise an exception with a clear and appropriate error message within the error-handling section. Ensure that all potential failure scenarios such as inserting a billing record for a non-existent patient or providing invalid billing values are properly handled and tested.

4. Designate one team member as the SQL DBA (database administrator) responsible for managing logins, users, and their privileges. Establish new roles within the database and assign appropriate privileges to each role. Then, based on the specific requirements of each user, assign relevant roles to them. Utilize the GRANT/REVOKE commands to control access to various objects within the database schema for each team member.
5. What backup strategy would you use for your database (e.g., full, differential, transaction log)? Please explain your backup and recovery plan, including what types of backups you would take and why.

Final Report (pdf)

The project report should include, but not limit to the following sections:

1. **Cover page**
Provide the title of the course, the title of the project, name of teacher, names of team members, and date.
2. **Table of contents**
Show the contents of the report and their corresponding page number.
3. **Introduction**
Provide a **brief** description of the project

4. **Conceptual Design of the Database**

The complete Entity-Relationship (ER) model of your database after all the modifications.

5. **Logical Database Schema**

- a. Give the schema of the database which is restructured and translated from the ER diagram presented in the section "Conceptual Design of the Database". Show the schema with appropriate referential constraints.
- b. Give the SQL statements used to construct the schema and the populated data

6. **Functional Dependencies and Database Normalization**

- a. Show the normalized tables for each relation to 3NF (if applicable).

7. The SQL statements for each of the defined queries and views, and its corresponding execution results as well as the stored procedures, trigger, controlling access statements, and backup strategy

Distribution of marks:

Topic	%
Changes in 2nd deliverable	5%
Queries Functionality	15%
Create Views	10%
Create Stored procedures	15%
Implement the Trigger	10%
Controlling user access	10%
Backup & Recovery	10%
Report Organization & Content	10%
Individual Contribution	15%
Total	100%